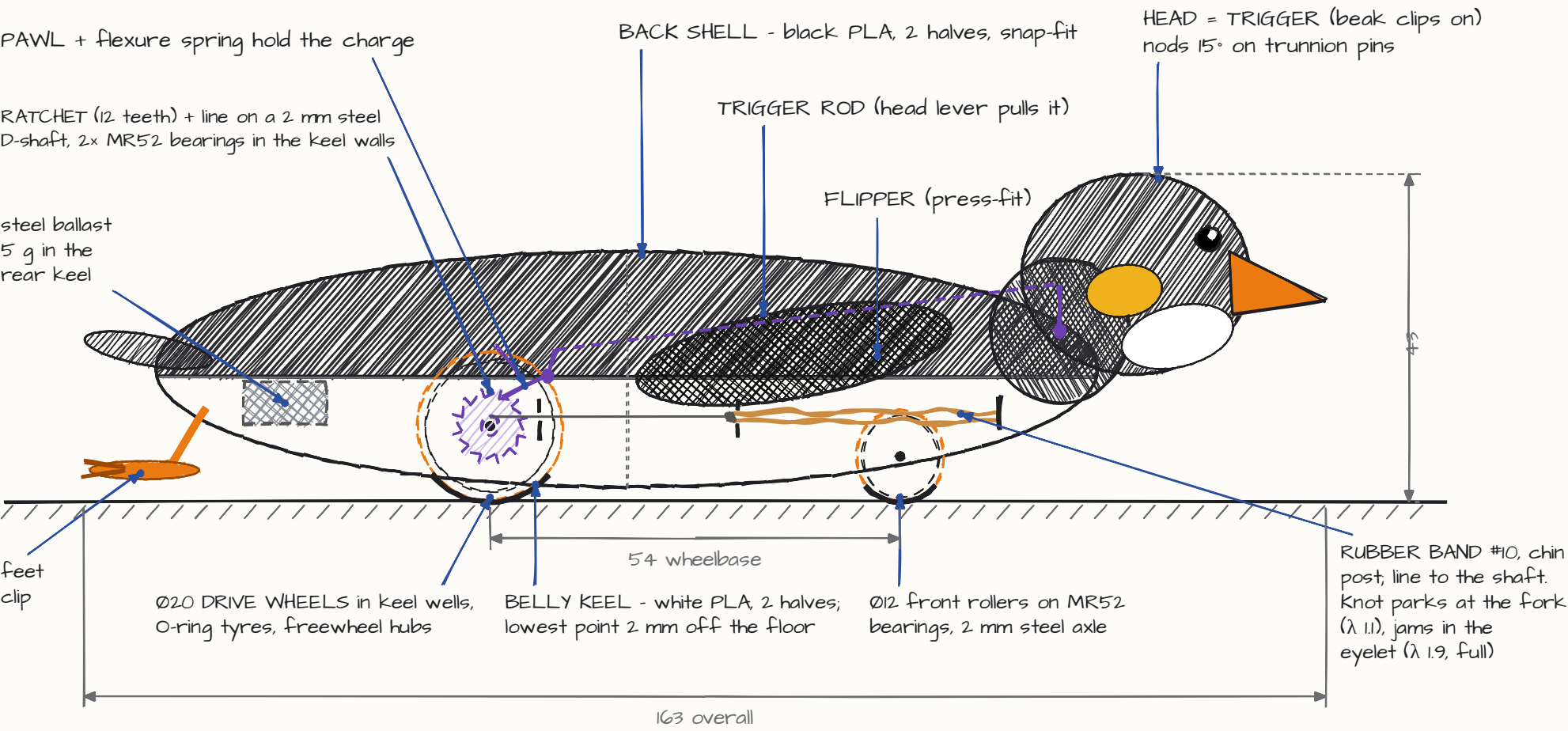
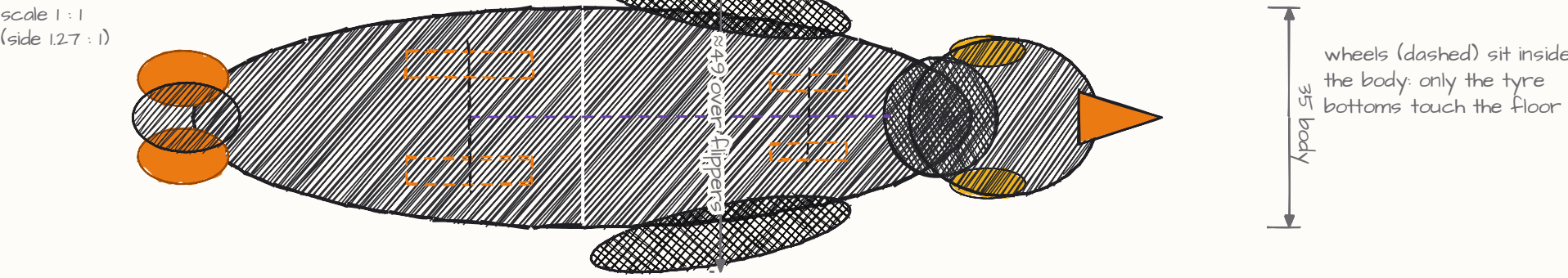


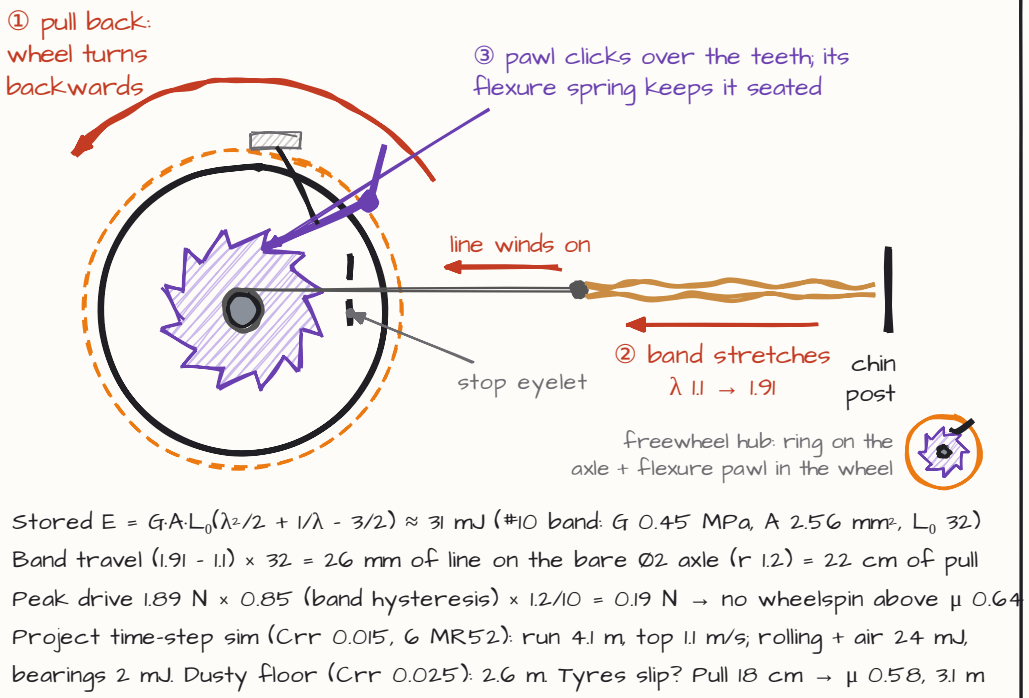
BODY - SIDE VIEW (scale 1:2.7 : 1 - dashed = hidden parts inside the body - purple = mechanism)



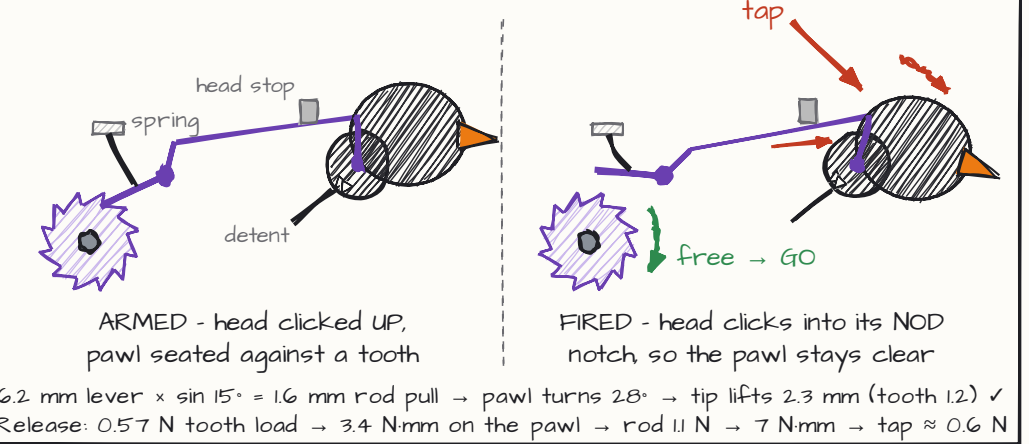
TOP VIEW



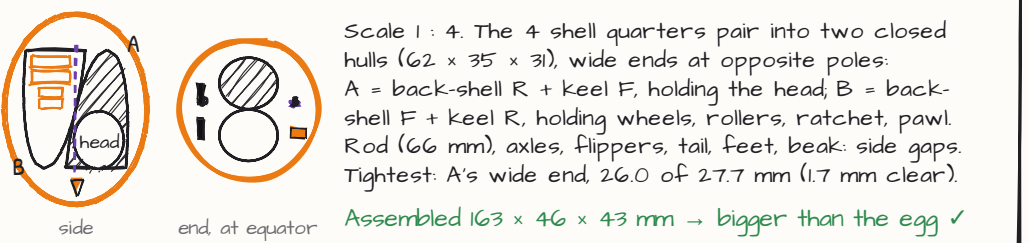
A ENERGY - PULL-BACK CHARGE



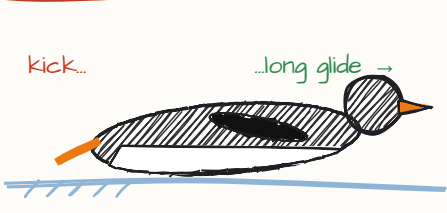
B TRIGGER - TAP THE HEAD



C PACKS INTO THE EGG



BIOMIMICRY



Tobogganing penguins (emperor, Adélie) give a few kicks, then glide far on their bellies. Copied: a 22 cm power stroke (the band), then a long freewheel belly-glide on a keel; low, long body.

HOW IT PLAYS

- Click the head UP, press on its back and pull back ~22 cm
 - Let go - the ratchet holds it.
 - Tap the head - it toboggans $\approx$ 4 m across the floor.
 - Lift the head to re-arm
- vs Concept 2: lying body, band in tension (pull-back), head-nod trigger.

NUMBERS

- Mass $\approx$ 39 g, 78 % printed PLA
- CG 17 mm ahead of rear axle, 15 mm up (68 % on drive wheels)
- Band stores 31 mJ; sim run 4.1 m (dusty-floor worst case 2.6 m)
- Top speed $\approx$ 1.1 m/s
- No wheelspin above μ 0.64 (nitrile O-ring $\approx$ 0.9; test it)
- Bought (15): 2 mm steel rod, 6x MR52, 4 O-rings, #10 band, line, 5 g ballast. Printed: 17 parts

RULES CHECK

- ✓ $\geq$ 75 % printed PLA (78 %)
- ✓ Parts fit the egg (box C); built toy is bigger than the egg
- ✓ Stores energy and waits until the head is tapped
- ✓ Trigger is not a removable pin - nothing comes off
- ✓ No energy stored when apart (band relaxed)
- ✓ Snap / press fits - no glue, no tools needed
- ✓ No bought toy parts or downloaded CAD

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 1 · sheet 1 of 1

Concept 1: TUX TOBOGGAN

Theme: Biomimicry - tobogganing penguin

Toy type: Running toy ($\geq$ 2.5 m)

Scale / units: mm · side view 1:2.7 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____